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Dear Professor Marc & Selection Team,

I am applying for the EngD on developing an **interactive reverse logistics map** for circular construction. The idea of a living map for reuse is what draws me in: a place where material information, logistics and decisions can come together so that good materials find a second life more often. When people cannot see or trust what is available, where it is, and how reuse compares on cost, CO_2 and circularity, new materials remain the safer choice. Helping build a map that changes that feels both useful and exciting.

During my master's studies, my thesis, **Agile Project Management in the Construction Industry**, explored how Agile can improve collaboration, continuous risk management, adaptive decision-making and resource efficiency in construction projects. Completing that work also made me realise that better project delivery and less construction waste address only one stage of sustainability. That led me to a broader question: what happens to materials when buildings reach the end of their service life? Looking into circular construction showed me how important reverse logistics is, and that a major barrier to reuse is not only technical feasibility. Information is often fragmented across the value chain. Details on material availability, quality, ownership, location, transport needs and reuse opportunities are incomplete or disconnected between stakeholders. Without reliable information, good materials may never return to the construction cycle.

Reading previous work in this field, including Professor Voordijk's research on recovering building elements for reuse, sharpened those questions for me. Reuse depends on coordination from the donor building to the target building, not only on technical innovation. That raises practical research questions for the map itself: how can material availability and condition be made visible across actors? How can donor and target needs be matched when timing, quality and cost are uncertain? How can a map support different stakeholders in comparing reuse options without adding another unused tool? These are the questions I am eager to work on, and they connect closely to the uncertain, volatile settings I studied in my thesis, where adaptive ways of working help teams respond when plans change.

My experience sits well with those questions. At Aug.e I worked on digital twins for smart buildings, using a digital environment to understand and improve how a real building performs. The reverse logistics map feels like a natural extension of that idea, applied to materials and their movement. At Hive CPQ I managed large manufacturer datasets and turned complex product information into simple digital solutions that non-technical users could work with. At iostring I worked with a construction partner on a circular construction hypothesis: how Agile project management can bring circular practices into real delivery. That work stayed at the hypothesis stage. Moving into a (de)construction pilot with partners such as Dura Vermeer and TNO is where I can contribute most: connecting construction knowledge, stakeholder needs and digital solution-building, supported by my civil and environmental engineering background and GIS skills. During a recent career break to care for my newborns, I kept learning about circular construction, reverse logistics and GIS, and I am ready to relocate to the Netherlands for this EngD.

What attracts me to the EngD at the University of Twente is the chance to grow through both advanced study and a real design project in industry. I want to deepen reverse-logistics modelling, learn to use GIS more rigorously for donor-target matching, and practise design-science research: define use cases with partners, test the map in a pilot, and improve it with the people who will use it. In return, I can bring energy, a construction-informed view of reuse, and hands-on experience translating messy project data into tools teams can actually use. Building this map is the kind of challenge I am keen to take on, and Twente's link between research and industry makes it an inspiring place to do that work.

Thank you for reading my application. I would be happy to discuss how I can contribute to the project.

Yours sincerely,
Pooja Awasare